exercise 1

Page → 3

p

page: 0 1 2 3
frame: 3, 5, 7, 0

$\frac{300}{256} = 1$ (Page)

Offset
300 mod 256 = 44

5 · 256 + 44 = 1324

300 entries[3] = 0 ← Frame

Page num: 300/256 = 3

Offset: 300 mod 256 = 32

Physical: 0 · 256 + 32 = 32

exercise 1, step 4

50 Frame = 3 valid

Page number: $\frac{50}{256} = 0$

Offset: 50 mod 256 = 50

Physical address:
3 · 256 + 50 = 768 + 50 = 818

1023

Page number: $\frac{1023}{256} = 3$

Offset: 1023 mod 256 = 255

Frame 0 ← valid

Physical address: 0 · 256 + 255 = 255

Va 1024 ← invalid

1024 ≥ 1024
not in valid range

Va
-1
-1 < 0

exercise 2

Process A:

Page: $\frac{300}{256} = 1$

Offset: 300 mod 256 = 44

A: entries[1] = {5, 1, 13, Use
Frame 5

Physical: 5 · 256 + 44 = 1324

Process B:

Page: $\frac{300}{256} = 1$

Offset: 300 mod 256 = 44

B: entries[1] = {2, 1, 13, Use
Frame 2

Physical: 2 · 256 + 44 = 556

600 after page load

Page number: $\frac{600}{256} = 2$

Offset: 600 mod 256 = 88

Frame lookup: {7, 1, 13}
↑ new!

Present in frame 7

Physical: 7 · 256 + 88
= 1792 + 88 = 1880